6 Explain the concept of arithemetic coding and illustrate its process with an example of compressing the string ABABABBB.

Arithemetic coding is a lossless - data compression technique used in data comprension systems. Unlike fixed length or variable-length codes, arithemetic coding represents an entire message as a single fractional number between 0 and 1.

Instead of assigning seperate bit codes to symbols, arithemetic coding progressively narrows a range [low, high] based on symbol probability until the final interval uniquely represents the message

The cumulative probabilities of symbols are represented on a line from 0.0 to 1.0. Each symbol occupies a subrange proportional to its probability.

New range = S + P(C) × R

S = cumulative probability of previous symbols

P(C) = Probability of current symbol

R = Current range (high - low)

Input string: ABAB ABBB

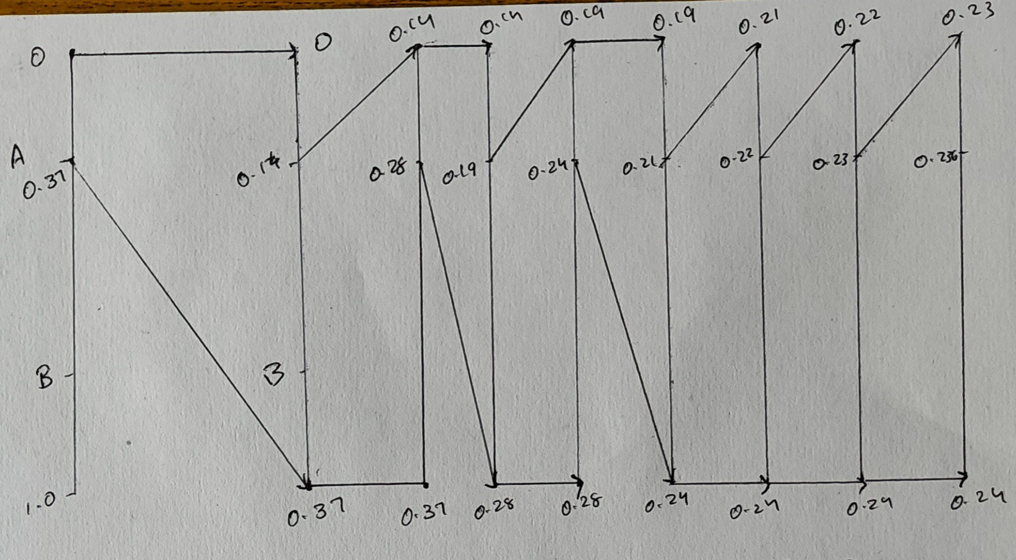
Number of A = 3

Number of B = 5

Probability of A = 3/8 = 0.37

Probability of B = 5/8 = 0.62

$F_1(A) = 0.37$ $F_1(B) = 1.0$.



i) $S + P(A) \times R$

$= 0 + 0.37 \times 0.37$

$= 0.14$

ii) $S + P(B) \times R$

$= 0.14 + 0.62 \times 0.23$

$= 0.28$

iii) $S + P(A) \times R$

$= 0.14 + 0.37 \times 0.14$

$= 0.19$

iv) $S + P(B) \times R$

$= 0.19 + 0.62 \times 0.09$

$= 0.24$

v) $S + P(A) \times R$

$= 0.19 + 0.37 \times 0.05$

$= 0.20$

vi) $S + P(B) \times R$

$= 0.21 + 0.62 \times 0.03$

$= 0.22$

vii) $S + P(B) \times R$

$= 0.22 + 0.62 \times 0.02$

$= 0.023$

viii) $S + P(B) \times R$

$= 0.23 + 0.62 \times 0.01$

$= 0.2362$

$Tag = \frac{0.23 + 0.24}{2} = \underline{\underline{0.235}}$

Arithemetic coding generates a unique tag for a sequence without building codes for all sequence of length m unlike huffmen coding.

The interval is repeatedly symbolized into the same proportions as the original range.

The advantages are high compression efficiency than huffmen coding and close to entropy limit.